CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 12: Reflection

Topic-wise short notes with solved examples

Reflection gives the mirror image of a point or a figure in a line or in a point. This chapter covers reflection in the axes, in the origin and in lines parallel to the axes, and the idea of invariant points.

1 Reflection in a line

- The **reflection (image)** of a point P in a line l is the point P' such that l is the perpendicular bisector of PP'.
- The line l is the **mirror line** (line of reflection or axis of reflection).
- P and P' are at equal distances from the mirror line, on opposite sides of it.
- The notation M_x means reflection in the x-axis, M_y reflection in the y-axis and M_o reflection in the origin.

2 Reflection in the axes and in the origin

Reflection in	Rule	What changes
The x-axis (M_x)	$(x, y) \to (x, -y)$	Sign of the ordinate
The y-axis (M_y)	$(x, y) \to (-x, y)$	Sign of the abscissa
The origin (M_o)	$(x, y) \to (-x, -y)$	Signs of both

Example 2.1 Find the reflection of $P(3, -4)$ in (i) the x-axis (ii) the y-axis (iii) the origin.

SOLUTION

(i) Change the sign of the ordinate: $(3, 4)$

(ii) Change the sign of the abscissa: $(-3, -4)$

(iii) Change both signs: $(-3, 4)$

(i) $(3, 4)$ **(ii)** $(-3, -4)$ **(iii)** $(-3, 4)$

Example 2.2 The point $P(2, 3)$ is reflected in the x-axis to P', and P' is reflected in the y-axis to P''. Find P' and P'', and name a single transformation that maps P to P''.

SOLUTION

$P' = (2, -3)$

$P'' = (-2, -3)$

$P'' = (-2, -3)$, **which is the reflection of P in the origin.**

REMEMBER

Reflection in the x-axis followed by reflection in the y-axis (in either order) is the same as reflection in the origin.

3 Reflection in a line parallel to an axis

Mirror line	Rule
$x = a$ (parallel to the y-axis)	$(x, y) \to (2a - x,\ y)$
$y = b$ (parallel to the x-axis)	$(x, y) \to (x,\ 2b - y)$

- The mirror line passes through the mid-point of the segment joining a point and its image. This fact finds the mirror line when a point and its image are known.

Example 3.1 Find the reflection of $P(5, 3)$ in (i) the line $x = 2$ (ii) the line $y = -1$.

SOLUTION

(i) $(2 \times 2 - 5,\ 3) = (-1, 3)$

(ii) $(5,\ 2 \times (-1) - 3) = (5, -5)$

(i) $(-1, 3)$ (ii) $(5, -5)$

Example 3.2 The image of $A(1, 2)$ in a line parallel to the y-axis is $A'(5, 2)$. Find the equation of the mirror line.

SOLUTION

The mirror line passes through the mid-point of AA', which is $(3, 2)$, and is parallel to the y-axis.

The mirror line is $x = 3$.

Example 3.3 A point P is reflected in the x-axis to $P'(4, -5)$. Find P, and the reflection of P in the line $y = 2$.

SOLUTION

Reflection in the x-axis changes the sign of the ordinate, so $P = (4, 5)$.

Reflection of P in $y = 2$: $(4,\ 2 \times 2 - 5) = (4, -1)$

$P = (4, 5)$; its reflection in $y = 2$ is $(4, -1)$.

4 Invariant points

- A point is **invariant** under a reflection if its image is the point itself.
- A point is invariant exactly when it lies **on the mirror line**.

Reflection in	Invariant points
The x-axis	Points on the x-axis, $(x, 0)$
The y-axis	Points on the y-axis, $(0, y)$
The origin	Only the origin $(0, 0)$
The line $x = a$	Points with abscissa a

Example 4.1 The point $(k, 0)$ is invariant under reflection in a line L, and so is $(0, 0)$. Name the line L.

SOLUTION

Both points lie on the x-axis, and invariant points lie on the mirror line.

L is the x-axis ($y = 0$).

5 Reflection of figures

- To reflect a figure, reflect each of its vertices and join the images in the same order.
- Reflection keeps lengths and angles unchanged, so a figure and its image are **congruent**. The orientation is reversed, as in a mirror.
- A figure together with its reflection is symmetrical about the mirror line.

Example 5.1 $P(3, 4)$ is reflected in the x-axis to Q; Q is reflected in the y-axis to R; and P is reflected in the y-axis to S. Write the coordinates of Q, R and S, name the figure $PQRS$ and find its area.

SOLUTION

$Q = (3, -4)$, $R = (-3, -4)$, $S = (-3, 4)$

$PQ = 8$ units (vertical) and $QR = 6$ units (horizontal), with right angles at each vertex.

$PQRS$ is a rectangle with area $8 \times 6 = 48$ square units.

6 Graph questions: how to answer

- Use the scale given, plot each point carefully and label it with its coordinates.
- Write the coordinates of every image asked for.
- Name the closed figure formed using the geometric name (kite, arrowhead, rectangle, isosceles trapezium, and so on).
- If asked, state the line of symmetry by its equation, such as $x = 0$ for the y-axis.

7 Common mistakes

- Changing the wrong sign: reflection in the x-axis changes y, not x.
- Using $(a - x, y)$ for reflection in $x = a$. The rule is $(2a - x, y)$.
- Calling a point invariant when it is not on the mirror line.
- Joining the image points in a different order from the original figure.

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